

ECON 6290: Labor Markets in Developing Countries

Week 4
Fall 2026

Lisa Ho

Announcements

1. Referee reports

- Thank you for submitting your second (RCT) referee reports, due today!

2. Mini-proposals

- Instructions have been posted

3. Shape of the day

- First half: finish fertility
- Second half: Saudi Arabia discussion, evolution of norms

4. Any questions?

Fertility and marriage

1. How worries about fertility flipped from over- to under-population

2. What changes marriage and fertility?

- **Jobs**

- **Jensen (2012)**, “Do Labor Market Opportunities Affect Young Women’s Work and Family Decisions? Experimental Evidence from India”, *Quarterly Journal of Economics*
- **Heath and Mobarak (2015)**, “Manufacturing Growth and the Lives of Bangladeshi Women”, *Journal of Development Economics*

- **Norms**

- **La Ferrara, Chong and Duryea (2012)**, “Soap Operas and Fertility: Evidence from Brazil”, *American Economic Journal: Applied Economics*

- **Agency: who decides?** (DHS data)

3. Policies to raise fertility?

Jobs I: BPO job recruiting in rural India (Jensen, 2012)

- Early 2000s: business process outsourcing (BPO) - e.g. call centers, data entry - on the rise
 - Starting pay Rs 5-10,000/month (US\$1-200), $\approx 2\times$ non-BPO pay at the same education
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- **Intervention:** experienced recruiters ran a 4-6 hr session for women in each treatment village
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 - "Booster" sessions one and two years later, plus three years of free placement support

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 - Described the job, how to apply, mock interviews, English assessment
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- **Data:** 20 random households/village (from full listing, no conditioning on attendance. why?)
 - Baseline survey: Sep-Oct 2003
 - Follow-up ("round 2") survey: Sep-Oct 2006

Result I: Effects of the intervention on employment

	BPO employment			Works for pay away from home		
	(1)	(2)	(3)	(4)	(5)	(6)
	18–24	25–44	45–60	18–24	25–44	45–60
<i>Panel A: Women</i>						
Treatment	0.046*** (0.008)	0.003 (0.003)	~	0.024** (0.011)	0.0029 (0.0089)	−0.006 (0.014)
Observations	1,278	2,233	1,029	1,278	2,233	1,029
Control group mean	0.004	0.002	0.00	0.21	0.24	0.22
R^2	0.022	0.000	~	0.054	0.001	0.000
<i>Panel B: Men</i>						
Treatment	−0.007 (0.005)	0.002 (0.004)	~	0.003 (0.011)	0.007 (0.024)	−0.004 (0.035)
Observations	1,442	2,469	1,104	1,442	2,469	1,104
Control group mean	0.008	0.003	0.00	0.47	0.56	0.52
R^2	0.001	0.000	~	0.000	0.001	0.000

Sample: all adults in sample HHs by age at round 2 (≈ 3 years after first recruiting session)

Result II: Effects of the intervention on young women's intentions to work

	(1) Control	(2) Treatment	(3) Difference (2) – (1)
Do you expect to work for pay (nonfamily) away from home . . .			
before you get married?	0.30 [0.46]	0.43 [0.49]	0.13** (0.051)
after marriage, but before you have children?	0.19 [0.40]	0.30 [0.46]	0.10** (0.048)
when your children are still young?	0.045 [0.21]	0.074 [0.27]	0.029 (0.026)
after all children have left school and are married?	0.23 [0.42]	0.34 [0.47]	0.11** (0.049)

Sample: women aged 18–24 at round 2 (≈ 3 years after first recruiting session)

Results III: Effects of the intervention on marriage and fertility

	(1) Married	(2) Had child	(3) Desired fertility
<i>Panel A: Women</i>			
Treatment	-0.051** (0.024)	-0.057** (0.026)	-0.35*** (0.078)
R^2	0.003	0.003	0.018
Observations	1,278	1,278	1,226
Control group mean	0.71	0.43	3.0
<i>Panel B: Men</i>			
Treatment	-0.002 (0.025)	-0.009 (0.018)	0.027 (0.066)
R^2	0.000	0.000	0.000
Observations	1,442	1,442	1,437
Control group mean	0.44	0.15	3.3

Sample: women and men aged 18–24 at round 2 (≈ 3 years after first recruiting session)

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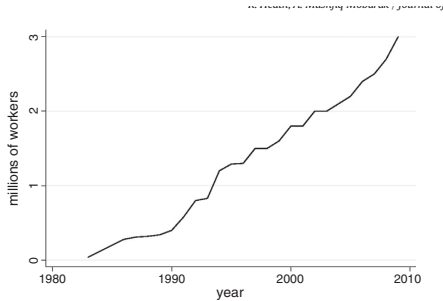
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Jobs II: Manufacturing growth and the lives of Bangladeshi women (Heath and Mobarak 2015)



Source: Bangladesh Garments Manufacturers and Exporter Association (BGMEA)

Fig. 2. Nation-wide employment in the garment industry.

Source: Bangladesh Garments Manufacturers and Exporter Association (BGMEA).

- **Garment industry:** 40,000 workers in 1983 → ≈ 4 million by 2013 (17% a year)
 - 79% of exports and 14% of GDP in 2008–09
- **Who gets hired:** industry estimates 80% of workers are women, and most are young women (20% are under age 18)
 - Hard to combine with marriage and children
 - Prefer those with education: garment workers have 6.0 years of school vs. 4.1 for others in their villages
- **Factory location** is important because most girls live in parents' house until marriage and can only take a factory job within commuting distance

Heath and Mobarak: data

- **Data:** authors' 2009 survey of 1,395 households in 60 peri-urban villages outside Dhaka
 - 44 villages within commuting distance of a factory vs 16 not
 - "First stage": 39% of women aged 16-30 in commute distance work in garment industry vs. 2% outside of commute distance
- **Sampling:** villages drawn at random from 4 subdistricts, stratified by commuting distance; census of each village, then random households (≈ 23 per village, more in larger villages; garment-worker households oversampled)

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- **Sampling:** villages drawn at random from 4 subdistricts, stratified by commuting distance; census of each village, then random households (≈ 23 per village, more in larger villages; garment-worker households oversampled)
- **Respondents:** HH head + wife report schooling, marriage, birth histories, work histories for all children, including those who have left home
- **One survey wave** (Aug-Sep 2009): the girl-year panel is retrospective (built from recalled ages)

Heath and Mobarak: empirical strategy

Variation: Villages are nearby garment work ($GV_v = 1$) or far ($GV_v = 0$)

- Work reached nearby villages in different years, so girls born in different years were exposed for different lengths of time
- Exposure is 0 in the 16 far villages, which pins down common time trends
- Specification lets nearby villages have their own linear trend

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Main specification (marriage; first birth is the same), for girl i in village v in year t :

$$\mathbb{1}(\text{Married})_{ivt} = \delta GV_v \times \text{YearsExposure}_{ivt} + \gamma_1 GV_v + \gamma_2 GV_v \times t + \beta \text{Age}_{ivt} + \lambda_t + \varepsilon_{ivt}$$

- YearsExposure: years since people in her village first worked in garments, from her birth to t
- δ : effect of one more year of garment-factory-exposure on the annual marriage hazard
- γ_1, γ_2 give the counterfactual in absence of exposure for nearby villages

Identification assumption?

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Identification assumption? Absent garment jobs, the **gap** in the marriage hazard between nearby and far villages would have kept moving along a straight line ($\gamma_1 + \gamma_2 t$)

Heath and Mobarak: results

Results:

1. Girls in garment-proximate villages **work outside the home** much more (15 pp more likely to have ever worked compared to 21.5%)
2. **Marriage is postponed:** with 6.4 years of exposure, 0.3pp decrease in marrying in a given year
 - Average marriage age was about 14
3. **First birth is postponed:** 0.23 pp lower chance of a first birth in a given year
 - Average age of first birth was about 19
4. **Schooling rises:** girls gain about **1.5 years** relative to their brothers
 - No effects for boys (makes roads and other investments an unlikely explanation)

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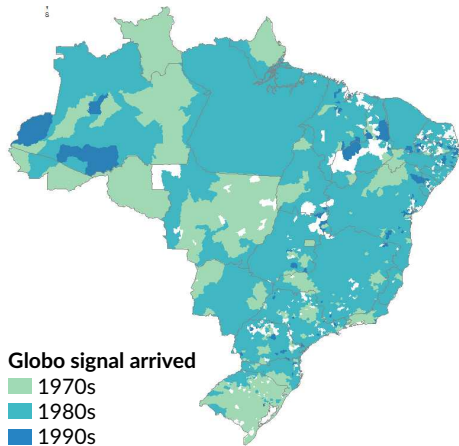
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Soap operas and fertility: evidence from Brazil (La Ferrara, Chong and Duryea 2012)



- **Novelas:** the 8 pm novela reaches 60–80 million viewers (they argue popular across classes)
- **Rede Globo:** near-monopoly on novelas; the military government's favored network from 1965
- **Small families on screen:** in 115 Globo novelas (1965–99), 72% of main female characters under 50 had **no children** and 21% had one
 - Most are unmarried; among married leads, 41% have none and 33% one
 - Per authors - family planning not explicitly addressed; plots featured women's emancipation, consumerism, upward mobility
- **Design:** staggered DID (area & year FE)
 - Variation: the year Globo's signal reached each of ~3,500 areas (1970, 1980, 1991 Censuses)

Soap operas and fertility: results

1. **Area level:** Globo coverage lowers average births per woman by **0.03** (mean 2.67)
2. **Individual level:** **0.6 pp** lower chance of giving birth in a year on a mean of **0.09** (about 6%)
 - The same as ≈ 2 more years of a woman's education where average education is 5.7 years
3. **Age:** no effect at ages 15–24; –8% of the mean at 25–34 and –11% at 35–44
 - Women stop earlier (not starting later); effects largest for poorer less educated HHs
4. **Television in general?** No because children are more often named after novela characters and other networks show no effect

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→ Norms about ideal family size may change fertility, too

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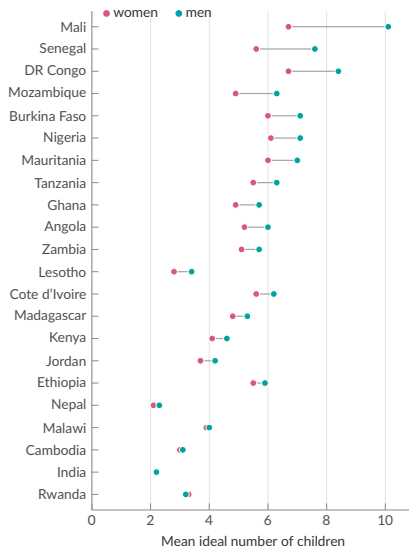
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Who decides: men and women want different numbers of children

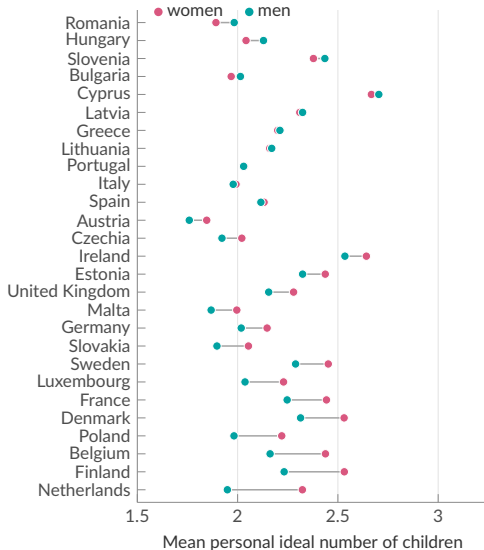
- Men report larger ideal family sizes than women in **62 of 69** countries with DHS data (generally LMICs)
- Maybe not surprising? Women bear the physical risks and costs of pregnancy and childbirth, most of the childcare, most of career costs
- So fertility is a bargaining outcome
 - Will depend on who decides when/whether to have sex and who controls contraception



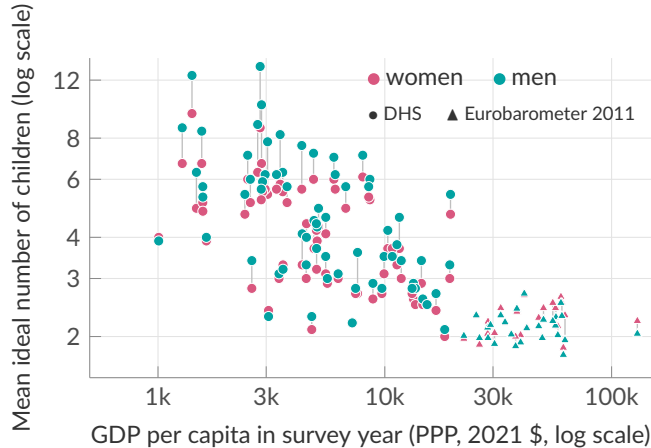
Mean ideal number of children, currently married women and men; 22 countries with data since 2020

That said... in Eurobarometer 2011, it is women who want (slightly) more children

- Eurobarometer 2011: “For you personally, what would be the ideal number of children you would like to have (or have liked to have had)?”
- Women’s personal ideal is higher than men’s in **19 of 27** EU countries
 - Average across countries: 2.23 for women vs. 2.14 for men
- Contrast with the DHS countries, where men want more in 62 of 69
 - Caveat that this is a decade earlier

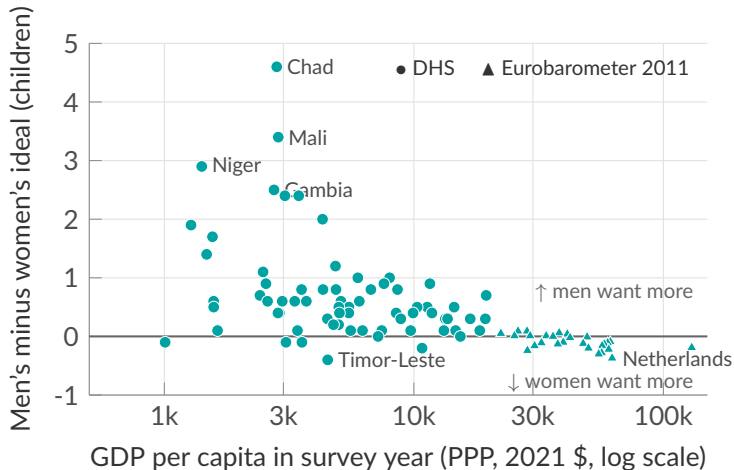


Ideal family size falls with income, for both men and women



DHS (69 countries, latest survey, currently married) and Eurobarometer 2011 (27 EU countries, ages 15–49); GDP per capita in survey year, World Bank.

The gender gap shrinks with income (and reverses in rich countries?)



Mean gap: +1.3 children below \$3k, +0.6 at \$3–10k, +0.3 at \$10–25k, –0.1 above \$25k

What lowers fertility?

- We've seen three forces (out of surely more) that lowered fertility and delayed marriage:
 - Better **opportunities** for women
Jensen (2012); Heath and Mobarak (2015); Baird, McIntosh and Özler (2011); Duflo, Dupas and Kremer (2015); Bandiera, Buehren, Burgess, Goldstein, Gulesci, Rasul and Sulaiman (2020)
 - Changing **norms** about family size
La Ferrara, Chong and Duryea (2012); Munshi and Myaux (2006); Jensen and Oster (2009); Kearney and Levine (2015)
 - More **say for women** in the decision
Rasul (2008); Ashraf, Field and Lee (2014); Doepke and Kindermann (2019)

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- Some in today's debate draw the conclusion: to raise fertility, reduce women's opportunities and decision-making power
- ... yikes !
- What about pro-natalist policies that have to do with childcare, parental leave, cash for births?

Fertility and marriage

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2. What changes marriage and fertility?
3. **Policies to raise fertility?**

What the causal evidence says about pronatalist policy

- **Cash for births:** baby bonuses and child benefits (Milligan 2005; Cohen, Dehejia and Romanov 2013; González 2013; Malkova 2018; Riphahn and Wijnck 2017; González and Trommlerová 2023; Kim 2024)
 - **Mostly +3 to +8% births/year** (Spain, Israel, Soviet Union; South Korea's local baby bonuses +2–5%); Quebec's large bonus for third-and-later children +12%; about zero in the US. Note this is all births per year, NOT completed family size

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 - Births per year: mostly **0 to +4%**; 0 in California, +4% in Germany
 - Completed fertility: mostly **0** (e.g. Norway, California); Austria: +3 births/100 mothers after 10 yrs
 - Leave for fathers lowered births in Spain? -7 to -15% within 2 years (Farré and González 2019)

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- **Childcare:** more childcare places or lower fees (Rindfuss, Guilkey, Morgan and Kravdal 2010; Bauernschuster, Hener and Rainer 2016; Mörk, Sjögren and Svaleryd 2013)
 - Births/year: **+3-10% births** per 10 pp of coverage: Germany, Norway; Sweden fee cap +10%
 - Completed fertility: **+0.1 to +0.7 children** per 10 pp of coverage: Germany, Norway

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 - Completed fertility: **+0.1 to +0.7 children** per 10 pp of coverage: Germany, Norway
- So some of these policies work, but effects are usually small and sometimes they don't work at all
 - And $\approx$ none of the effect sizes are of the magnitude to get to replacement rate
- Kind of a trendy area to study, but difficult to get credible variation and measure *completed* fertility
- Unclear what implications are for welfare, or what fertility rate is “good” and for whom

Norms, beliefs, and attitudes

1. **Discussion: Bursztyn, González and Yanagizawa-Drott (2020)**, “Misperceived Social Norms: Women Working Outside the Home in Saudi Arabia”, *American Economic Review* (discussion lead: Elena)
2. **When do norms change? Gulesci, Jindani, La Ferrara, Smerdon, Sulaiman and Young (2025)**, “A Stepping Stone Approach to Norm Transitions”, *American Economic Review*

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A Stepping Stone Approach to Norm Transitions

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- Approach taken by governments and NGOs is often to push for the outright abandonment of norms seen as harmful (e.g. child marriage, female genital cutting)
 - Often, though, this approach is ineffective
- For the purpose of eliminating a harmful norm in the long run, does the existence of a “mildly harmful” alternative facilitate the transition compared to a binary choice scenario?
 - Examples include cap on dowry payments, a lower minimum age for marriage than the eventual age you want to reach, or a less invasive form of female genital cutting (FGC)...

Stepping stones: why might they help or hurt?

Why they might help:

- People who may be reluctant to completely abandon a harmful practice may be persuaded instead to adopt a less harmful variant
 - Once this first step is taken, it may be easier to take further steps and eventually eliminate the practice altogether

Why they might hurt:

- On the other hand, precisely because the variant is less costly, incentives to abandon it are lower, and it may become a new absorbing norm

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This paper studies norms around female genital cutting (FGC), and two types of cutting:

1. **Pharaonic** or “infibulation”: labia cut and joined, often stitched; historically near-universal in Somalia; often undone then redone for intercourse/childbirth; most severe health risks
2. **Sunna**: partial or total removal of the clitoris, sometimes the labia minora; spread as religious leaders called Pharaonic harmful and non-Islamic but endorsed Sunna; still painful/health risks

Type of FGC by Year of Cutting

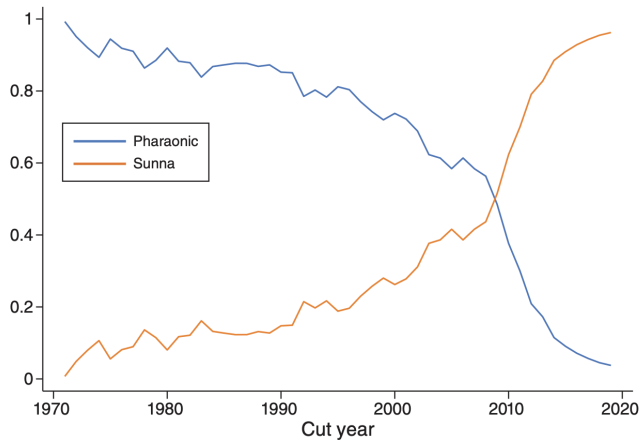


FIGURE 1. TYPE OF FGC BY YEAR OF CUTTING

The paper's model of social norms

- Build on Akerlof (1997) and Brock and Durlauf (2001)
- Agents choose an action to maximize a utility that has two components:
 1. the **intrinsic utility** of the action for the individual
 2. a **social utility** that is decreasing in the shares of agents who choose actions different from the individual's
- Agents' utilities are subject to shocks, leading to heterogeneity in individual choice

Model Set Up: Part 1

- Population of people who can choose from actions L, M, and H (for low, medium, and high intrinsic utility, respectively). Define $A = \{L, M, H\}$
 - In the case of FGC in Somalia, L represents Pharaonic, M represents Sunna, and H represents not cutting (the ranking reflects the health costs of each action)

Model Set Up: Part 1

- Population of people who can choose from actions L, M, and H (for low, medium, and high intrinsic utility, respectively). Define $A = \{L, M, H\}$
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- Time is continuous, and agents revise their actions via independent Poisson arrival processes

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- Departure from most of literature: social pressure doesn't need to be symmetric
 - i.e. allow for $s_{ij} \neq s_{ji}$
 - For example, agents who choose costly actions may greatly disapprove of agents choosing uncostly actions, but agents who chose uncostly actions might exert little social pressure on other agents

Model Set Up: Part 3

- When an agent revises at time t , her utility for action i is subject to a shock ϵ_{it} :
$$u_i - \sum_{j \in A} s_{ji} p_{jt} + \epsilon_{it}$$
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- Extreme-value shocks lead to logit choice probabilities
 - Probability of agent choosing action i in state p is $\sigma_i(p) = \frac{e^{\beta v_i(p)}}{\sum_{j \in A} e^{\beta v_j(p)}}$
 - $\beta > 0$ is agents' precision and inversely related to variance of the shocks

When β is small, agents' choices almost random

When β large, agents' choices close to pure best response

Model dynamics

- Shocks incorporate the idea that people are heterogeneous with respect to their intrinsic utilities and also with respect to the social pressure coefficients
- Because of this heterogeneity, at any given point in time, the prevailing norm will be followed by most people but not by everyone
- When the population is large, the expected direction of motion is given by $\sigma(p)$, which is the flow conditional on the current stock p
- One can think of p as a stock and $\sigma(p)$ as a flow

Definition: Stepping Stone

M is a stepping stone if:

Starting from $p(0) = p^L$ (everyone chooses L):

- 1. The flow switches from M to H:** $\exists t' > 0$ such that
 - $\sigma_M(p(t)) > \sigma_H(p(t))$ for $t \in [0, t')$ (initial dominance of M)
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One implication of (7a) and (7c) is “reverse triangle inequality:”

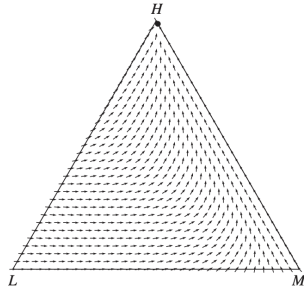
$$s_{LH} > s_{LM} + s_{MH} \quad (1)$$

Direct jump is punished more than the two steps put together, i.e. sanctions have to be convex:

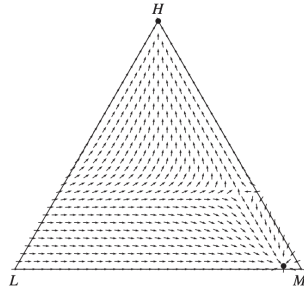
- a small departure from the norm is tolerated
- a big departure is punished more than proportionally

4 Possible Dynamics

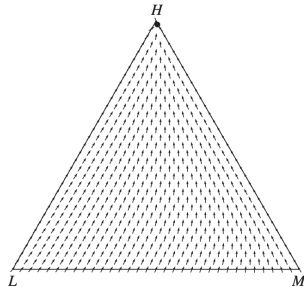
Panel A. M is a stepping stone



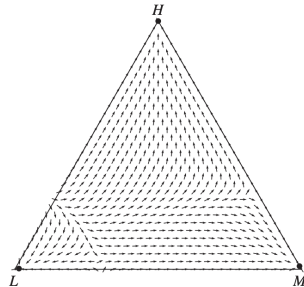
Panel B. M is absorbing



Panel C. Direct transition to H



Panel D. L is absorbing



Empirical application: FGC in Somalia

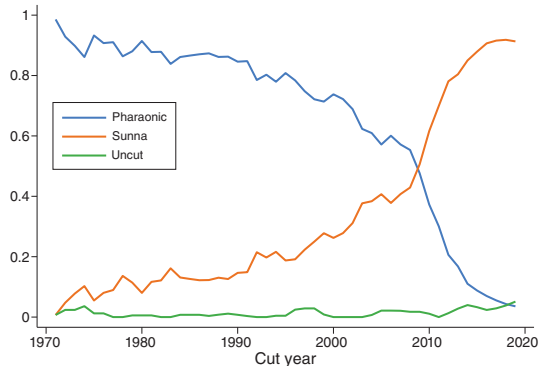
- **Question:** Pharaonic (*L*) has largely given way to Sunna (*M*). Is Sunna a stepping stone to not cutting (*H*) or is it an absorbing state?
- **Survey**, January–May 2020: 141 communities in Somaliland and Puntland, 4,130 respondents (2,040 men, 2,090 women)
 - Cutting type of each woman and each of her daughters, including the planned type for girls not yet cut
 - The mother made the decision for about 80% of cut girls
- Women and girls aged 10–60 as of survey: **33% Pharaonic, 64% Sunna, 3% uncut**

Turning the survey into the model's stock and flow

- **Stock** $p(t)$: shares of women and girls aged 10+ in year t who are Pharaonic, Sunna, uncut
- **Flow** $\sigma(t)$: shares of girls *turning 10* in year t with each status — the choice made by that year's “revising” parents
 - Average age at cutting is 9; planned cuts count as cut
- **Flows (right)**: Sunna overtakes Pharaonic around 2010; uncut ticks up slightly

Three-year moving averages; cohorts born after 1960.

Flows $\sigma(t)$, 1970–2020



Use logit to turn model into a regression

- From the logit choice probabilities, the ratio of two flows depends only on the difference in utilities:

$$\frac{\sigma_H(p)}{\sigma_M(p)} = \frac{e^{\beta v_H(p)}}{e^{\beta v_M(p)}} \Rightarrow \ln \frac{\sigma_H(p)}{\sigma_M(p)} = \beta [v_H(p) - v_M(p)]$$

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- Plug in $v_i(p) = u_i - \sum_j s_{ji} p_j$ and use $p_M = 1 - p_L - p_H$:

$$\ln \frac{\sigma_H}{\sigma_M} = \underbrace{\beta(u_H - u_M - s_{MH})}_{\alpha} + \underbrace{\beta(s_{HM} + s_{MH})}_{\gamma > 0} p_H + \beta(s_{LM} + s_{MH} - s_{LH}) p_L$$

- Left side: observed flows. Right side: observed stocks.
- Estimated by OLS with one observation per year (or district-year)

Test 1: the reverse triangle inequality

- The coefficient on the Pharaonic stock p_L is $\beta (s_{LM} + s_{MH} - s_{LH})$, and $\beta > 0$. So:
 - Negative $\Leftrightarrow s_{LH} > s_{LM} + s_{MH}$: the inequality holds
 - Positive $\Leftrightarrow$ it fails, so Sunna cannot be a stepping stone

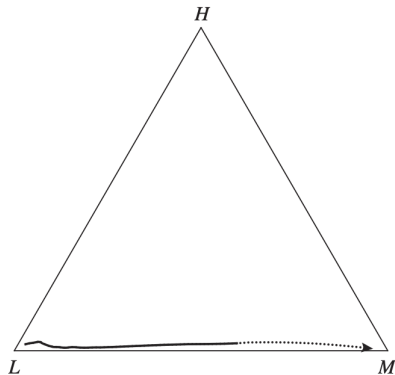
Dep. var.: $\ln(\sigma_H/\sigma_M)$	(1)	(2)	(3)	(4)	(5)
Pharaonic stock p_L	3.70 (0.58)	5.13 (0.66)	5.75 (0.65)	4.86 (0.87)	5.27 (0.86)
Uncut stock p_H	109.7 (18.4)	35.1 (7.5)	36.6 (13.9)	72.0 (16.8)	72.8 (15.1)
Observations	Year	Year-district	+ district FE	Year-clan	+ clan FE

- Positive and significant everywhere: inequality fails, evidence against stepping stone

Test 2: extrapolation

- The path depends on the parameters only through $\sigma(p)$, estimate that directly
- **Step 1:** regress $\ln(\sigma_H/\sigma_M)$ and $\ln(\sigma_M/\sigma_L)$ on p_H and p_L . With $\sigma_L + \sigma_M + \sigma_H = 1$, that pins down all three flows
- **Step 2:** start from today's stock and let the stock move toward the flow, year after year
- **Result:** the path heads to (nearly) all Sunna. The recent rise in uncut continues for a while, then reverses

Panel A. Extrapolation test



Solid: data. Dotted: simulated.

Test 3: equal flow

- If Sunna is a stepping stone, the process must pass the inflection point, where the flows into H and M are equal
- Check directly community by community:
$$\frac{\sigma_H}{\sigma_H + \sigma_M} \geq \frac{1}{2}?$$
 - Girls who reached cutting age in 2010–2020
- Find not a single community has reached $\frac{1}{2}$
- Consistent with Sunna being absorbing in the absence of a shock

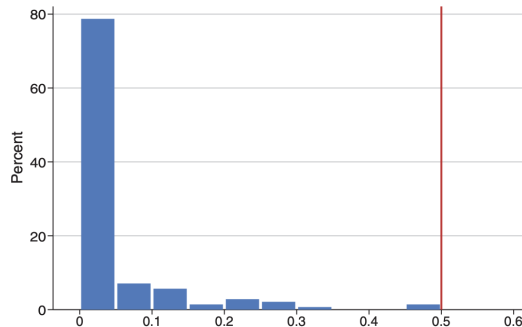


FIGURE 9. FREQUENCY OF $\sigma_H / (\sigma_H + \sigma_M)$ VALUES ACROSS COMMUNITIES

Framework applies to many norms:

Paper uses dueling as a historical example

- UK: Direct abolition ($L \rightarrow H$)
- France: Shift to less lethal forms, then gradual decline ($L \rightarrow M \rightarrow H$)

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Other examples they mention:

- **Female labor force participation:** Home $\rightarrow$ Remote work $\rightarrow$ Office?
- **Child marriage:** Very young $\rightarrow$ Teen marriage $\rightarrow$ Adult marriage?
- **Smoking:** Cigarettes $\rightarrow$ E-cigarettes $\rightarrow$ Quitting?

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Questions for any context:

1. Do social sanctions satisfy reverse triangle inequality?
2. Is intermediate option sufficiently costly to motivate further change?